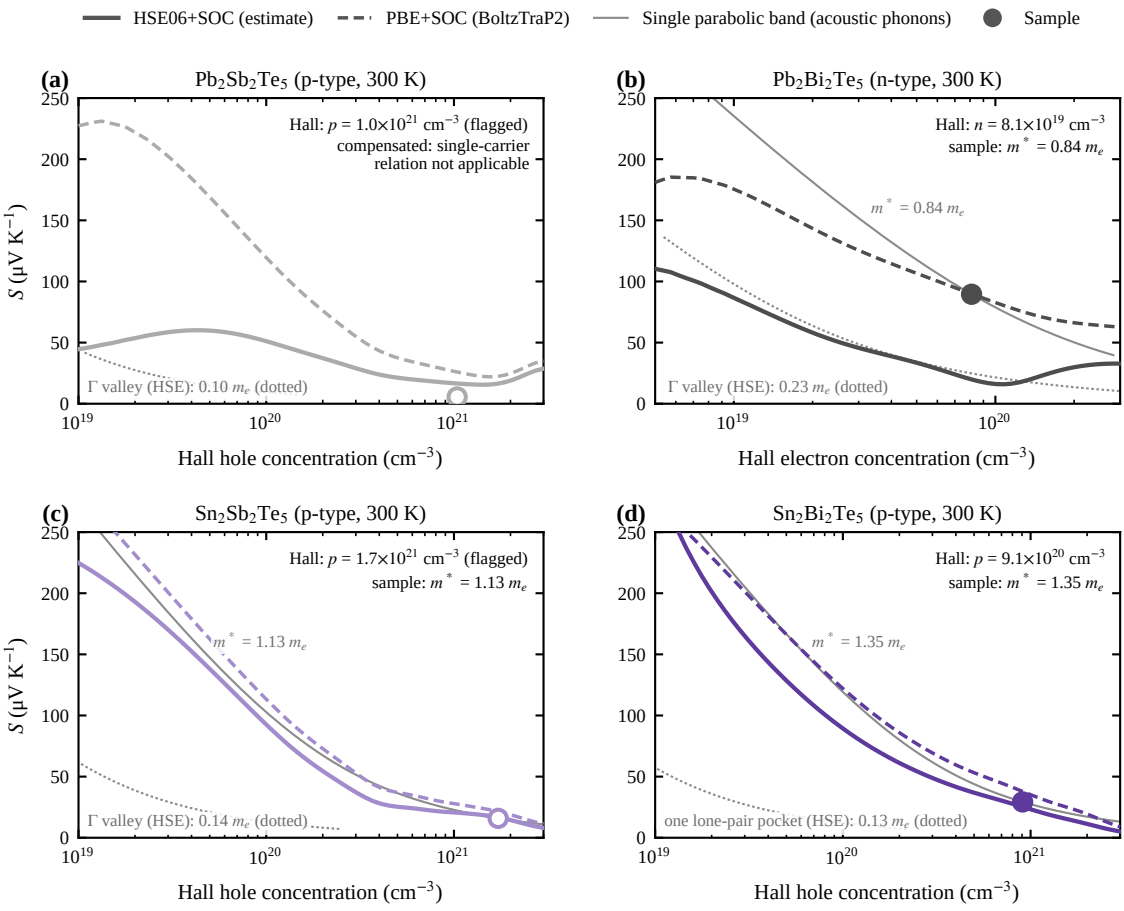
HSE06+SOC (estimate)
PBE+SOC (BoltzTraP2)
Single parabolic band (acoustic phonons)
Sample
(a)
Pb2Sb2Te5 (p-type, 300 K)
Hall: p = 1.0×10^21 cm^−3 (flagged)
compensated: single-carrier relation not applicable
Γ valley (HSE): 0.10 m_e (dotted)
S (μV K^−1)
Hall hole concentration (cm^−3)
(b)
Pb2Bi2Te5 (n-type, 300 K)
Hall: n = 8.1×10^19 cm^−3
sample: m* = 0.84 m_e
m* = 0.84 m_e
Γ valley (HSE): 0.23 m_e (dotted)
Hall electron concentration (cm^−3)
(c)
Sn2Sb2Te5 (p-type, 300 K)
Hall: p = 1.7×10^21 cm^−3 (flagged)
sample: m* = 1.13 m_e
m* = 1.13 m_e
Γ valley (HSE): 0.14 m_e (dotted)
Hall hole concentration (cm^−3)
(d)
Sn2Bi2Te5 (p-type, 300 K)
Hall: p = 9.1×10^20 cm^−3
sample: m* = 1.35 m_e
m* = 1.35 m_e
one lone-pair pocket (HSE): 0.13 m_e (dotted)
Hall hole concentration (cm^−3)